

Let's learn about **satellites!**

Space mission and **satellite** design course

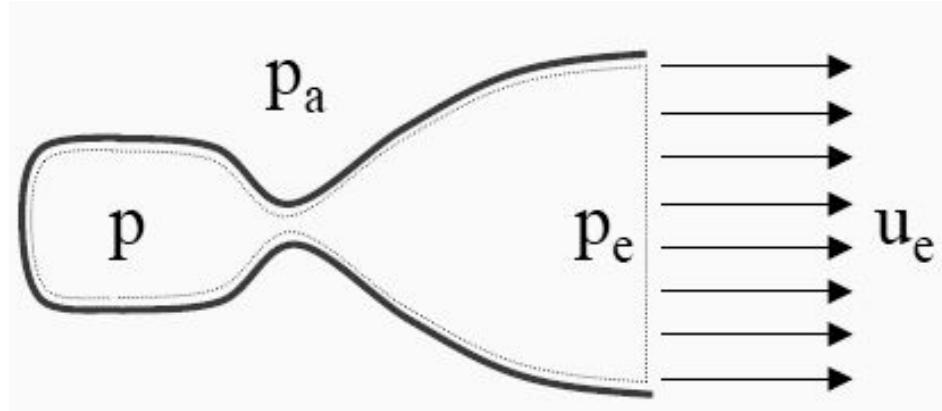
Satellite propulsion

What is a propulsion system?

- A propulsion system is a subsystem that generate a **thrust**
- Chemical thrusters produce thrust by exerting an equal and opposite force on a fluid (air, combustion products, etc.)



How does propulsion works?



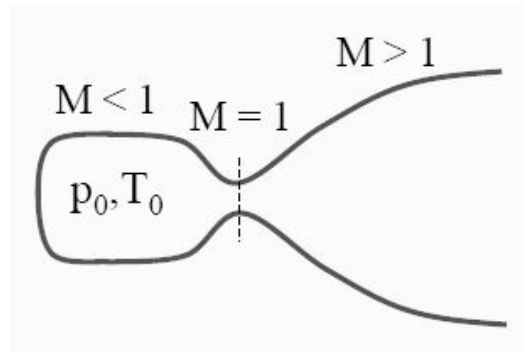
$$F = m * a = \dot{m}_e * u_e + (p_e - p_a) * A_e$$

What is M?

- The **Mach** number is a dimensionless number defined as the ratio between the speed of an object moving in a fluid and the speed of sound in the fluid in question.
 - When $M=1$ the velocity is equal to the sound speed
 - When $M>1$ the velocity is higher than the sound speed (Supersonic condition)
 - When $M<1$ the velocity is lower than the sound speed (Subsonic condition)

How is made?

- Chemical thrusters typically consist of a high-pressure combustion chamber and a convergent-divergent nozzle
- Typically the flow is subsonic in the converging section, sonic in the throat and supersonic in the diverging section of the nozzle
- The speed of the flow at the throat is not influenced by conditions downstream («choked flow»)



Specific impulse

- Specific impulse measures the impulse produced per unit of propellant and is proportional to the exhaust velocity.
- it is an index that expresses how effectively the energy content of the propellant is converted into thrust.

$$I_{sp} = \frac{u_e}{g_0}$$

<u>Type</u>	<u>Specific impulse (sec)</u>
Cold gas	50-75
Solid rocket motors	280-300
Liquid monopropellant	150-225
Liquid bipropellant	300-450

Classifications

Propulsion systems are typically used for:

- Launch of payloads into LEO orbits
- Payload transfer from LEO up to mission orbits (higher orbits or on interplanetary trajectories)
- Provide thrust for orbit maintenance (station keeping) and attitude control
- Obtain spacecraft orientation via attitude control
- De-orbit or re-enter the vehicle

System requirements and performance depend on the application (large launch vehicles or medium-sized propulsion)

Classifications



Same concept... different size!

Classifications

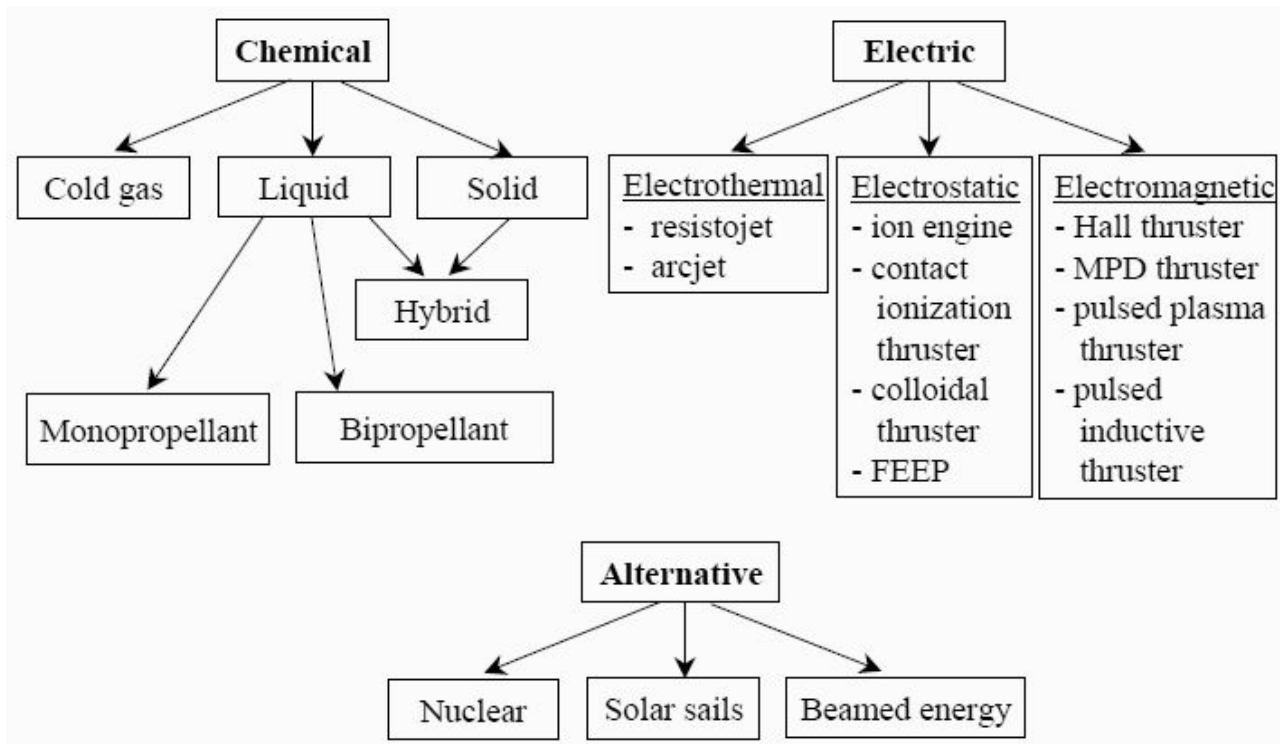
- Selecting the best propulsion system for a given mission is complex and depends primarily on performance criteria, such as thrust requirements, total impulse, duty cycle, and speed variation.
- Other requirements not strictly related to system performance include operating pressure and internal or external losses, position of thrusters, type of propellant, weight.

Classifications

Propulsion systems are classified based on the type of energy source and are typically divided into 3 categories:

- **Chemicals:** The energy that produces thrust is contained in the propellant, which is released by chemical reactions; following which the propellant is accelerated to high speeds, expanding in gaseous form through a nozzle.
- **Electric:** the energy that produces thrust is not contained in the propellant but must be supplied externally by an auxiliary power source, such as nuclear energy, solar radiation or batteries. The thrust in this case is provided by the expansion of hot gases (heated by electric current) or by the acceleration of charged particles in electric or magnetic fields
- **Alternative** («exotic»)

Classifications



Classifications

- The main methods of space propulsion are based on the principle of "jet propulsion", in which thrust is produced by the expulsion of the matter that was contained within the system, called propellant.
- The thrust generated will depend on the exit speed of this material, i.e. on the acceleration that it has been possible to impart to the combustion products

Classifications

- Spacecraft propulsion systems are characterized by complete integration within the bus and are designed to include less propellant and produce lower levels of thrust than those of launch vehicles.
- The propulsion system of the launchers is defined as "**primary**", while that on board the satellites is "**secondary**".



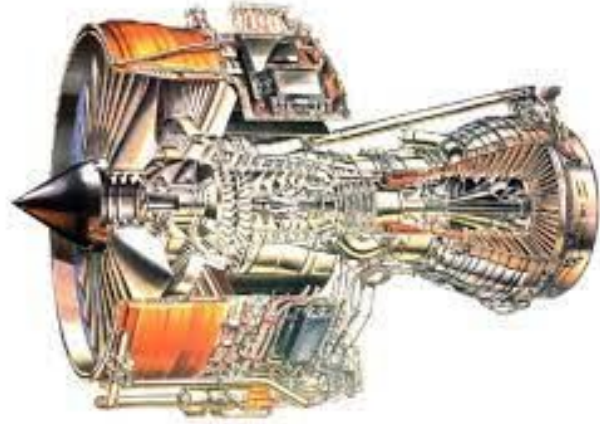
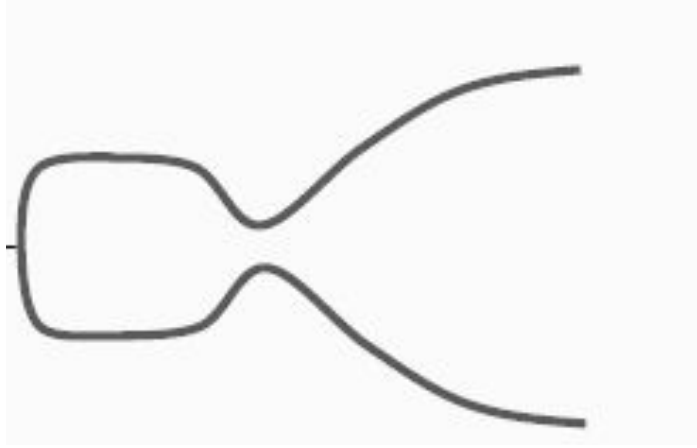
Chemical propulsion

Chemical propulsion is based on the principle of conversion of chemical energy into kinetic energy of exhaust gases inside a nozzle of a propulsion system

Chemical thruster architectures generally include the following elements:

- Combustion chamber: where the propellant burns, producing hot gas
- Converging section, "throat", to control chamber pressure and mass flow
- A nozzle, with a divergent section, to accelerate the gas
- Propellant storage and supply systems
- Valves and pipes that connect the tanks with the thrusters
- Electrical control unit of valves and thrusters

Chemical propulsion



Chemical propulsion



Cold gas propulsion

Operates with propellants such as inert compressed gases, hydrogen N_2 , or high pressure hydrocarbon vapors, propane C_3H_8

They represent a low-performance, economical solution, which are mainly used when there are specific requirements to avoid hot gases and safety concerns related to the handling of liquid or solid propellants.

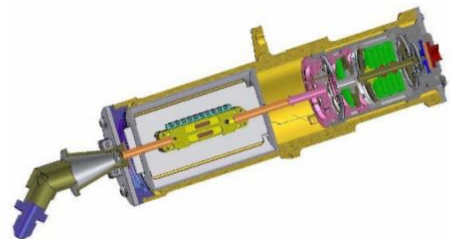
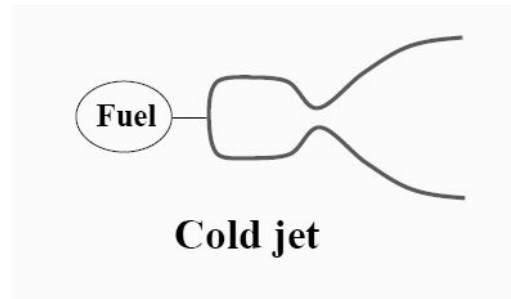
- Simple design: solenoid valve + nozzle
- Simple operation: activating the valve releases the gas flow
- The thrust levels achieved depend on the inlet gas pressure and the nozzle diameter
- Often used for attitude control

Cold gas propulsion

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Chemical propulsion

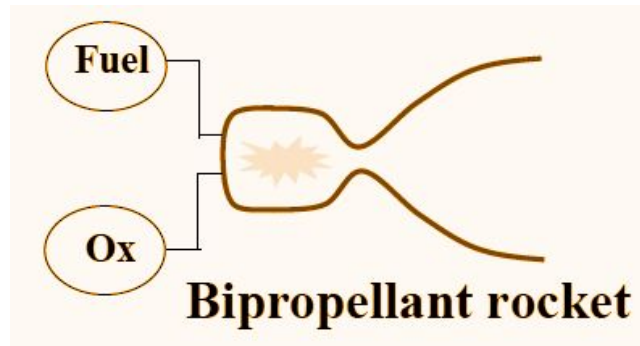
- If we want combustion to increase energy we need the condition to have it
- **Combustion** occurs thanks to an oxidizer (e.g. oxygen) and a fuel (e.g. petrol)
- On earth we just care about fuel cause we have air, but how do we do in space?



Bi-propellant thrusters

We bring oxidizer with us!

- They operate by burning two propellants to produce thrust, i.e. one that acts as a fuel and one as an oxidizer.
- The propellants are injected separately into the combustion chamber where they react spontaneously to obtain high temperatures and generate low molecular weight combustion products, which are expelled through a nozzle.

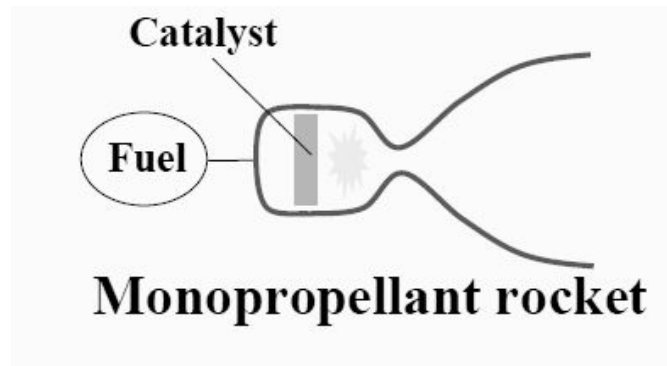


Bi-propellant thrusters

Engine	Supplier	Nominal Thrust (N)	Specific Impulse (sec)	Chamber Pressure (atm)	Engine Mass (kg)	Propellant O/F	Burn time (sec)	Flight status
XLR-132	Rocketdyne	1.67×10^4	340	102.0	51	N ₂ O ₂ /MMH	4000	in development
R40-A	Marquardt	3.87×10^3	306	10.5	10	N ₂ O ₂ /MMH	15,320	Shuttle OC
LR-105-5	Rocketdyne	3.86×10^5	316	48.0	460	LOX/Ker.	430	Atlas
RS-27A	Rocketdyne	1.05×10^6	302	48.0	1,091	LOX/Ker.	274	Delta
RL-50	P&W	1.10×10^5	450	102.0	317	LOX/LOH	630	design concept, Centaur
SSME	Rocketdyne	2.28×10^6	455	204.0	3,177	LOX/LOH	480	Space Shuttle
LEROS 1c	ARC	460	324	11.0	4.26	N ₂ O ₂ /MMH	29,000	LM spacecraft
LEROS 10	ARC	9.5	285	6.4	0.46	N ₂ O ₂ /MMH	N/A	in development

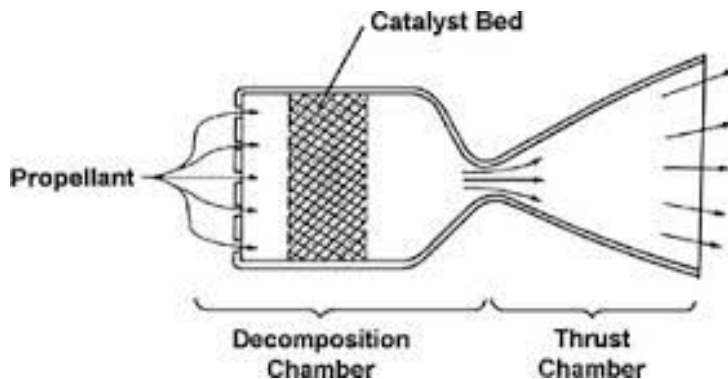
Mono-propellant thrusters

- They operate on only one type of propellant to produce thrust.
- The most commonly used monopropellant is hydrazine (N_2H_4)
- They are often used for attitude control and station-keeping
- The chemical reaction that releases energy occurs thanks to the interaction between hydrazine and a **catalyst**



Mono-propellant thrusters

- Hydrogen peroxide, also known as hydrogen peroxide, is the simplest of the peroxides. Its chemical formula is H_2O_2
- Hydrogen peroxide decomposes into water and oxygen; this reaction is very slow but can be accelerated by a catalyst!



Mono-propellant thrusters

Engine	Supplier	Nominal Thrust (N)	Specific Impulse (sec)	Chamber pressure (atm)	Engine Mass (kg)	Min. Impulse (N sec)	Max. Impulse (N sec)	Flight status
MR-103G	Aerojet	0.2-1.1	202-224	4.5-23.8	0.33	0.013	90,188	Iridium, A2100
MR-104A,C	Aerojet	205-573	223-239	3.9-10.7	1.86	8.23	693,900	DMSP, Landsat
MR-106E	Aerojet	11.6-30.7	229-235	4.5-12.4	0.52	0.46	120,000	Echostar, A2100
MR-111C	Aerojet	1.3-5.3	215-229	3.4-12.1	0.33	0.08	260,000	Intelsat, Telstar
MONARC 1	AMPAC	1.0	232	7-28	0.33	0.011	111,250	qualified
MONARC 5	AMPAC	5.0	230	5-30	0.37	0.09	311,500	qualified
MONARC 90	AMPAC	90.0	235	5-31	1.00	0.89	3,500,000	qualified

Electrical propulsion

- To increase the performance of the propulsion system in terms of impulse to meet the demands of, for example, **interplanetary missions**, the exhaust velocity must grow above 5000 m/s, which is the maximum that can be achieved with chemical propulsion.
- This can be achieved with electric propulsion!
 - **Electrothermal systems** (resistojet and arc-jet), in which the expansion of hot gases (heated by electric current) takes place inside a nozzle
 - **Electromagnetic systems** (MPD motors), in which a plasma is accelerated thanks to the interaction of electric and magnetic fields
 - **Electrostatic systems** (ion engines), in which the acceleration of charged particles in an electric field occurs

Energy can be obtained from sunlight, via solar panels, or from nuclear reactors.

Electrical propulsion

Electric propulsion systems include the following main components:

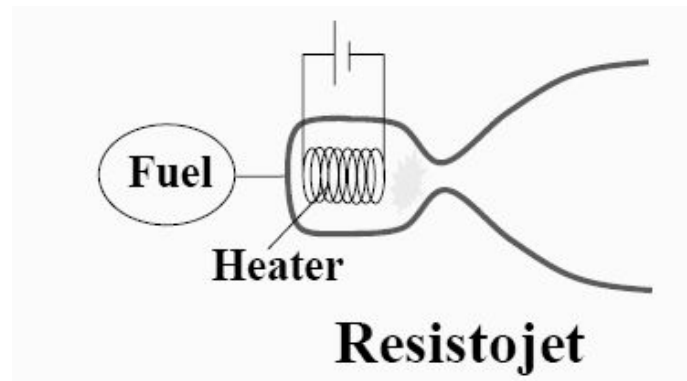
- Storage and supply system that supplies propellant to the thruster
- Valves and pipes to connect the tank to the thruster
- Electrical control unit for valves and thrusters
- Electrical energy source and power conditioning system

Energy processing systems are needed to convert the voltage from the electrical power generator to the level required by the thruster. For example, the typical voltage level of an ion thruster is on the order of **kilovolts**, which is very far from the voltage provided by solar panels or typical EPS,

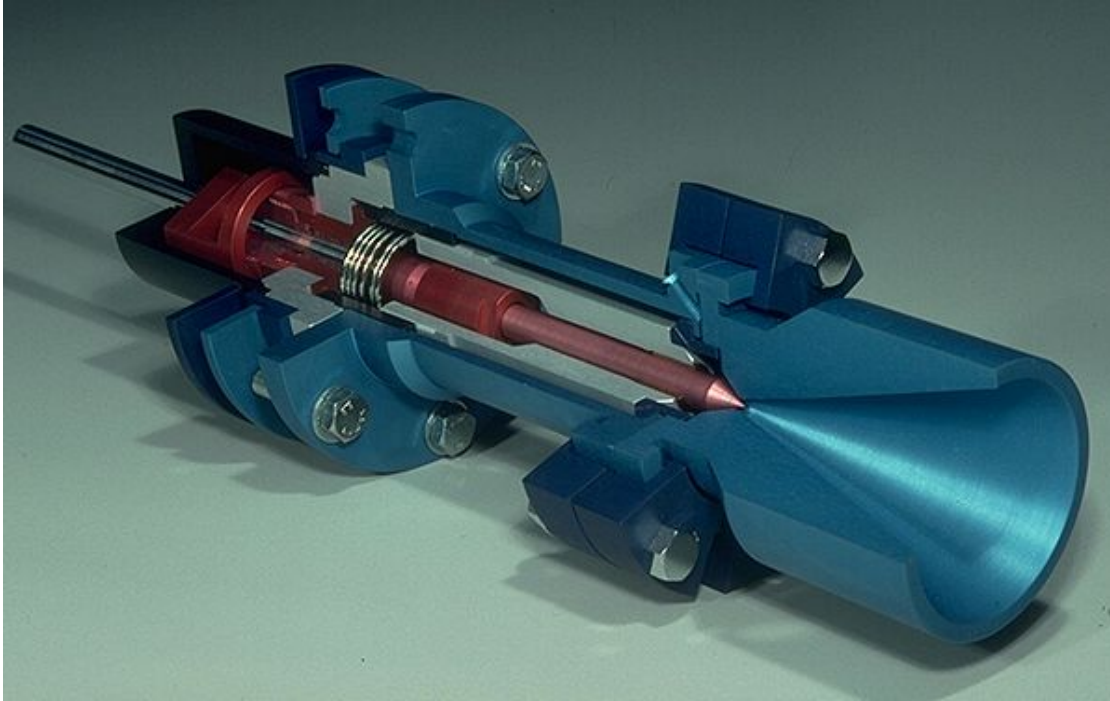
Electrical propulsion

Resistojet

- This is a so-called electrothermal system, where the propellant, gas, is heated by passing over an element heated with resistive electric heaters.
- The heated gas is then accelerated inside a gas expansion nozzle.
- The resistojet is the simplest electrothermal thruster.
- The most used propellants are hydrogen, nitrogen, ammonia, or the most common, hydrazine (PACT engines)



Electrical propulsion

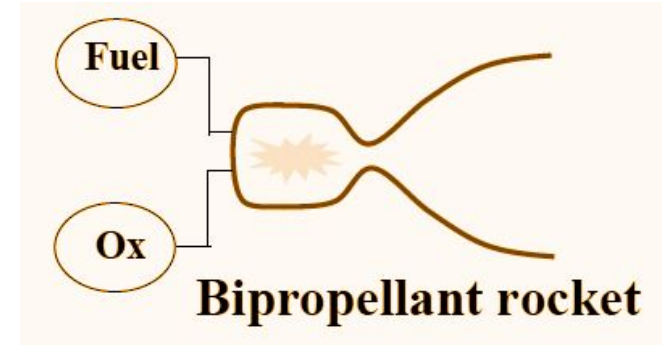
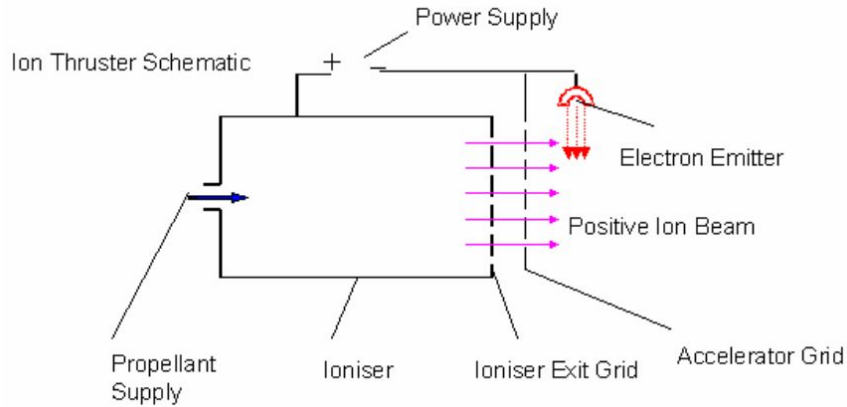


Electrical propulsion

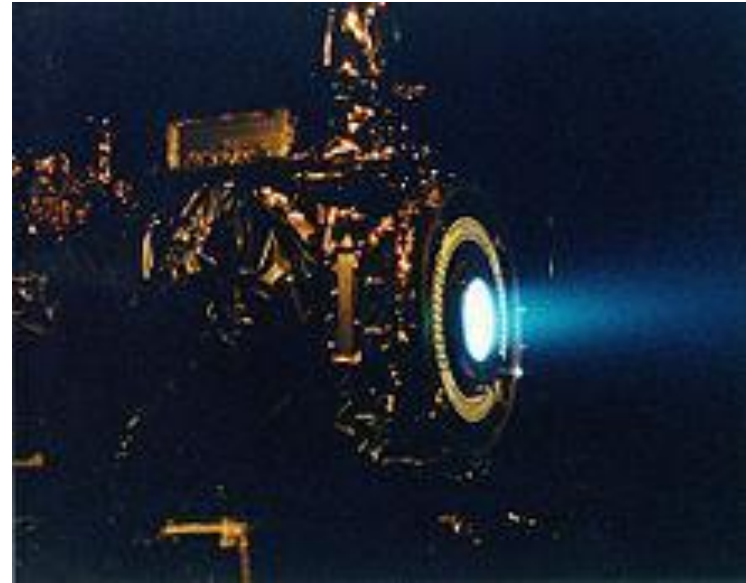
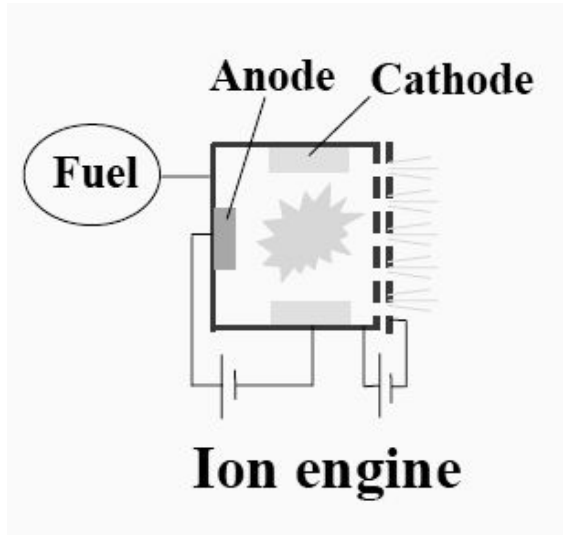
Electrostatic Propulsion

- A propellant (xenon gas) is ionized, through electronic bombardment, in a high frequency electromagnetic field (radio frequencies).
- The generated ions are then accelerated to high speeds (between 30 and 60 km/s) via a strong electric field.
- The electrons are then injected into the ion beam from an emitter to keep it electrically neutral, preventing the generation of a differential charge on the spacecraft
- They are generically called **ion engines**
- Among electric motors they represent the highest performance solutions
-

Electrical propulsion



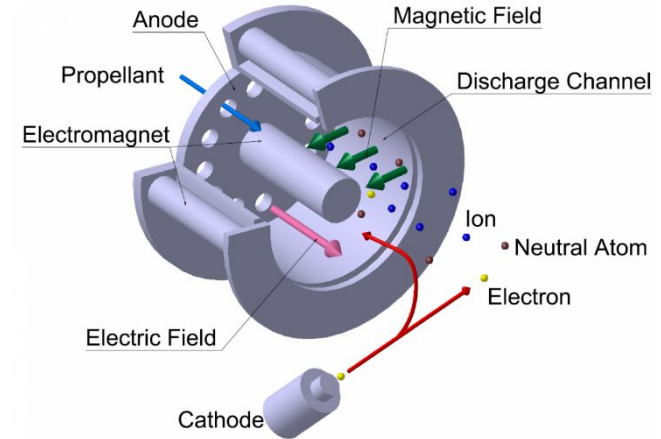
Electrical propulsion



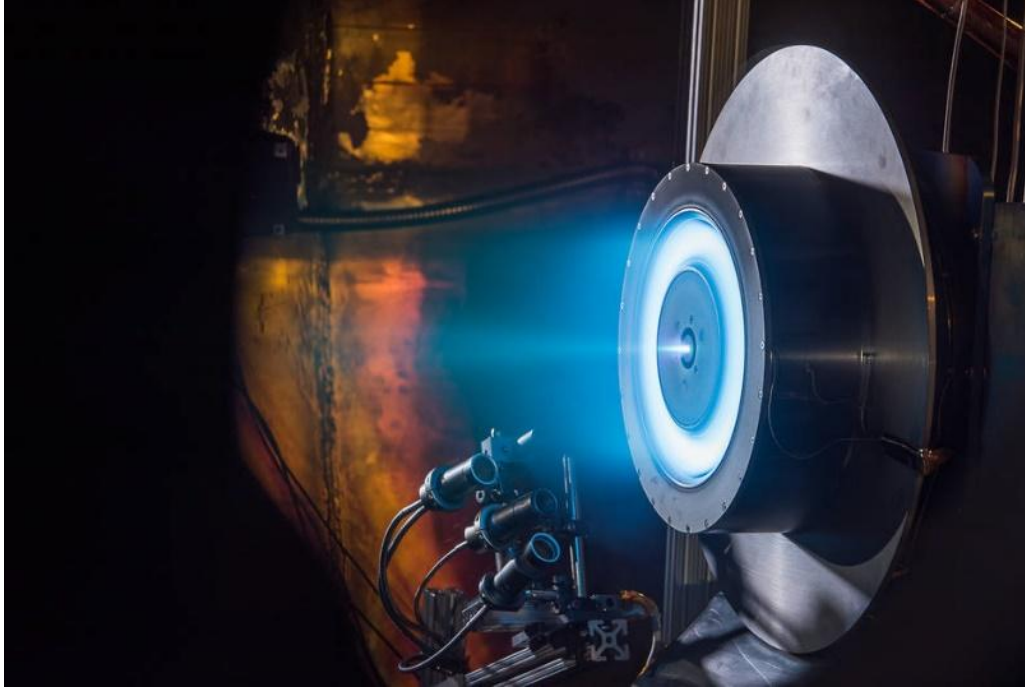
Electrical propulsion

Hall effect motors

- In these motors a radial magnetic field is required.
- Within the volume of this magnetic field a continuous axial electrical discharge is maintained in the propellant gas (xenon) at low pressure.
- The interaction between the radial and axial fields generates a Hall current, perpendicular to the electric field.
- This Hall electric current interacts with the magnetic field and generates an axial force, which results in an axial acceleration of the propellant.



Electrical propulsion



Electrical propulsion

Hall effect motors

- In these systems a gas is heated through an electric arc to such a high temperature that it is converted into a neutral plasma.
- The plasma is then expelled at high speed by the interaction of the electric discharge current and a magnetic field, generating Lorentz forces.
- These engines are called plasma thrusters, and the most common model is the Magneto-Plasma-Dynamic (MPD) thruster.

